温州肯恩大学
WENZHOU-KEAN UNIVERSITY

MGS 3101 W03 Foundations of Business Analytics

Statistical Inference II: Interval Estimation

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Week 9-2, 30 Oct 2025

Midterm Exam Review

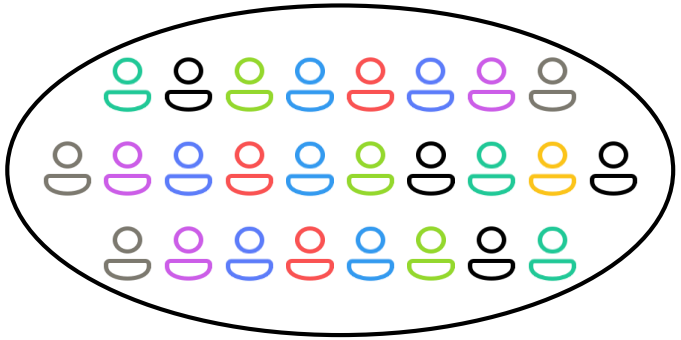
OR

- Average: 86.83
- Common Issues & Key Points
  - (1c) Ratio scale/data: order value in RMB (revenue per transaction) **vs.** day of the week the purchase occurred
  - (2c) Explain what the skewness implies about customer spending behavior, not just statistical terms
  - (2e) The number of mobile orders is higher **vs.** The average revenue per mobile order is higher
  - (2f) Missing calculation of covariance
  - (2g) Strength & direction; -0.067 indicates very weak negative linear association/relationship
  - (3a-3b): Some errors in using functions (e.g., invalid arguments for `plt.hist()` or `plt.bar()`)
- Reminder
  - If you have any questions about your grades, comments, or specific parts of the exam, please contact me or visit during office hours. Grades will be finalized on **Nov 6**. No changes after this date.

Review of Last Class

Point Estimation
- Estimator vs. estimate
- What is a good estimator?

A Population (N)

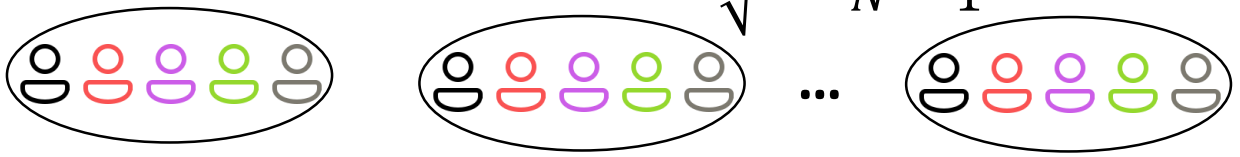


What
Why
How

Sampling

Inference

A Sample (n)



sample 1 $(\bar{x}_1, s_1)$ sample 2 $(\bar{x}_2, s_2)$ … sample m $(\bar{x}_n, s_n)$

Population parameter
- Mean: $\boldsymbol{\mu}$
- Standard deviation: $\boldsymbol{\sigma}$

$$\mu = \frac{\sum_{i=1}^{N} x_i}{N}$$

$$\sigma = \sqrt{\frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N-1}}$$

Sample estimate
- Mean: $\bar{\boldsymbol{x}}$
- Standard deviation: $\boldsymbol{s}$

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n} \qquad \bar{x} = 15$$

$$s = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}} \qquad s = 3$$

Sampling distribution of sample mean $\bar{\boldsymbol{x}}$, a random variable, point estimator of μ

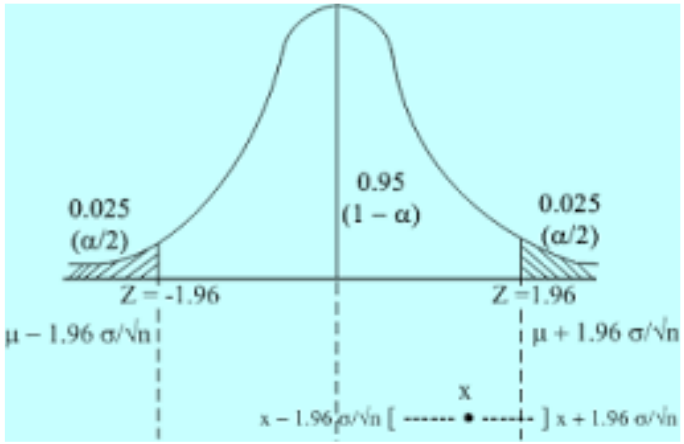
CLT
- $E[\bar{\boldsymbol{x}}] = \mu$, unbiased estimator; $\sigma_{\bar{x}} = \sigma/\sqrt{n}$ if population is infinite or $n/N < 0.05$.
- Regardless of distribution of population, when <u>*sample size n is large enough*</u>, $\sqrt{n}(\bar{x} - \mu) \xrightarrow{d} N(0, \sigma^2)$
- If the population is normally distributed, then the sampling distribution is also normally distributed for <u>*any* sample size</u>.

Interval Estimation

- What
  - An interval estimate provides a range of values that is likely to include the population parameter.
  - Intervals specify a range of plausible values for the characteristic of interest and a way of assessing "how plausible" they are.
- How

$$\text{Interval Estimate} = \text{Point Estimate} \pm \text{Margin of Error}$$

  - The margin of error depends on:
    - The **variability** of the estimator (<u>standard error</u>)
    - The **confidence level** (e.g., <u>95%</u>)



Margin of Error = Critical Value * Std

- Example
  - In a normal distribution, 90% (95%) of the values lies within 1.645 (1.960) standard deviations of the mean.

$$\mu \pm 1.645\sigma \qquad \mu \pm 1.960\sigma$$

  - Thus, when sampling distribution $\bar{x}$ is normal, 90% (95%) of the values lies within 1.645 (1.960) standard deviations ($\boldsymbol{\sigma_{\bar{x}}}$) of the mean.

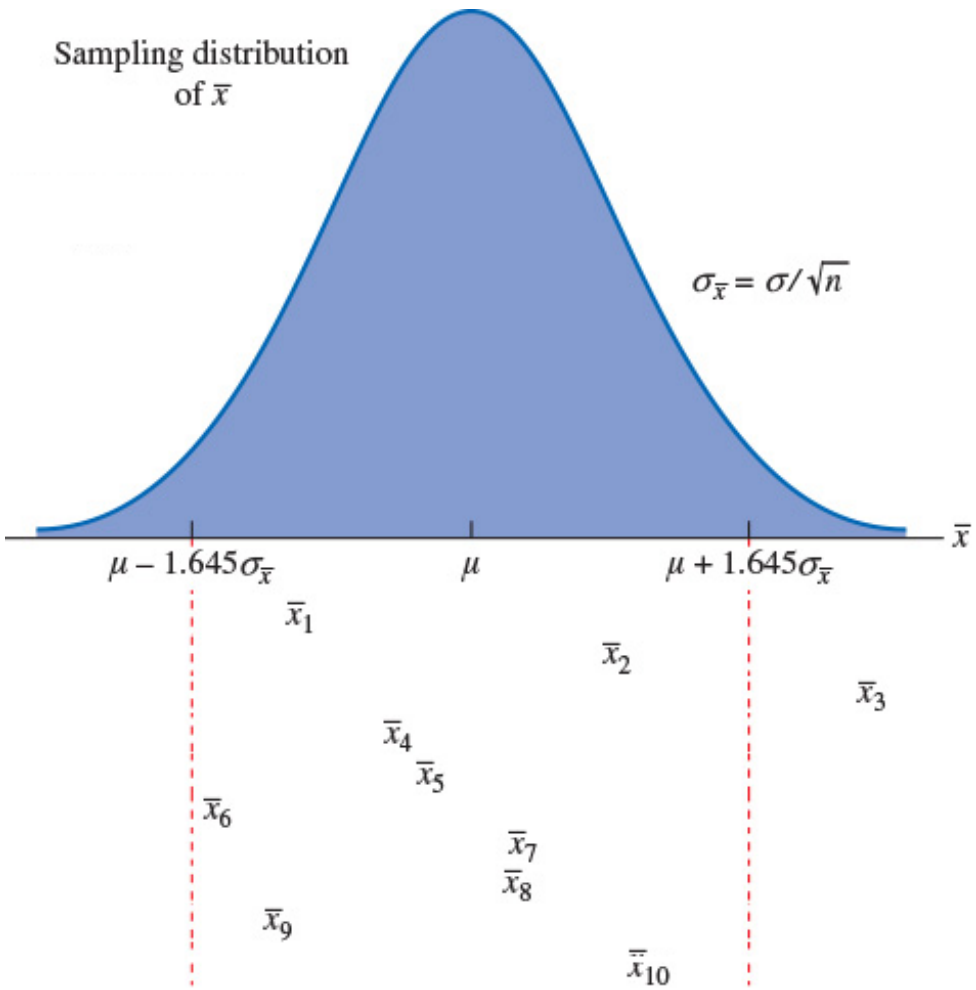
$$E[\bar{x}] \pm 1.645\sigma_{\bar{x}} \qquad E[\bar{x}] \pm 1.960\sigma_{\bar{x}}$$

Sampling Distribution of $\bar{x}$

- Consider a sampling distribution of $\bar{x}$ with

  $\boldsymbol{E(\bar{x}) = \mu}$ and $\boldsymbol{\sigma_{\bar{x}} = \sigma/\sqrt{n}}$

- If the sampling distribution follows the normal model, it can be shown that
  - 90% of $\bar{x}$ values lie within $1.645\sigma_{\bar{x}}$ of μ
  - 95% of $\bar{x}$ values lie within $1.960\sigma_{\bar{x}}$ of μ
  - 99% of $\bar{x}$ values lie within $2.576\sigma_{\bar{x}}$ of μ
- *<u>We use the sampling distribution of $\bar{x}$ to produce interval estimates of the population mean μ.</u>*



Confidence Level

- Because approximately 90% of all the intervals constructed will contain the population mean μ we say that

  "We are approximately 90% confident that the interval will include the population mean μ, and that the interval has been established at the 90% confidence level."

- The value of 0.90 is referred to as the confidence coefficient.
- The interval estimate is called the 90% confidence interval.
- The level of significance is *the probability that the interval estimation procedure will generate an interval that does not contain the population mean*.

$$\alpha = \text{level of significance} = 1 - \text{confidence coefficient}$$

Remember: $\sigma_{\bar{x}} = \sigma/\sqrt{n}$

Under what condition?

Confident Interval with Known Population $\boldsymbol{\sigma}$

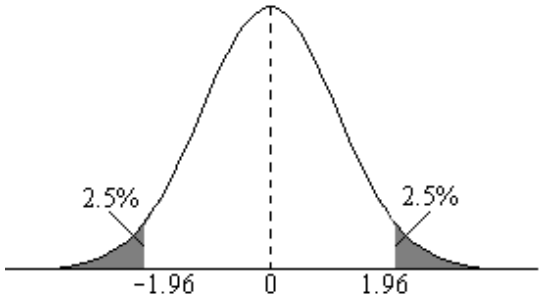
Interval Estimation of a Population Mean: $\boldsymbol{\bar{x} \pm z_{\alpha/2}(\sigma/\sqrt{n})}$

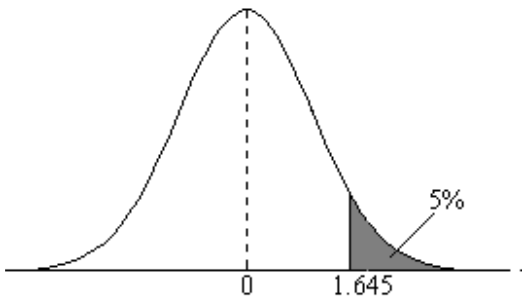
- By CTL, we have $\sqrt{n}(\bar{x} - \mu) \xrightarrow{d} N(0, \sigma^2)$ OR $\dfrac{\bar{x} - \mu}{\boldsymbol{\sigma}/\sqrt{n}} \xrightarrow{d} N(0,1)$

z score, z test

z-distribution = standard normal distribution

- Sample mean $\pm$ margin of error
- Margin of error is: $\boldsymbol{z_{\alpha/2}(standard\ error)}$
  - $\boldsymbol{z_{\alpha/2}}$: value of standard normal random variable for an upper tail area of $z_{\alpha/2}$ (or a lower tail area of $1 - z_{\alpha/2}$.
  - Examples
    - If $\alpha = 0.05$, for a 95% confidence interval, then $z_{0.05/2} = 1.96$.
    - If $\alpha = 0.10$, for a 90% confidence interval, then $z_{0.10/2} = 1.645$.





Confident Interval with Unknown Population $\boldsymbol{\sigma}$

Interval Estimation of a Population Mean: $\bar{\boldsymbol{x}} \pm \boldsymbol{t}_{\alpha/2}(\boldsymbol{s}/\sqrt{\boldsymbol{n}})$

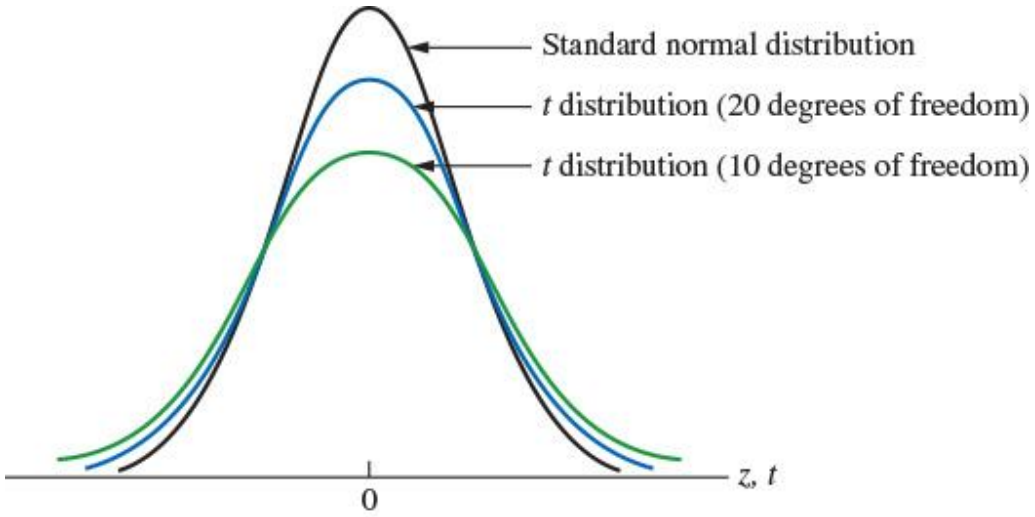
- But, the population σ is often unknown, we use the sample standard deviation s to estimate it.
- This leads to a new statistic: $$t = \frac{\bar{x} - \mu}{\boldsymbol{s}/\sqrt{n}} \sim t_{n-1}$$ **t score, t test**

  which follows *a t-distribution with ($n - 1$) degrees of freedom*.
- The **sample mean** $\bar{x}$ itself still follows a **normal distribution** (approximately, by CLT) — but because σ is replaced by s, we use the *t-distribution* to make inference.
- Using the *sample standard deviation (s)* to replace the unknown *population standard deviation (σ)* introduces additional uncertainty.
- This uncertainty is accounted for by the *t-distribution* — a *family of distributions* whose exact shape depends on the *degrees of freedom (df = n − 1).*

This uncertainty/error is related to sample size n.

The Family of t Distribution

- The t-distribution is defined by **only one** parameter: *degree of freedom = no. of sample values – no. of estimated parameters*
- These t-distributions are similar in shape to the *standard normal distribution*, but wider.
- As the degrees of freedom (sample size n) increase, the t distribution narrows, its peak becomes higher, and it becomes more similar to the standard normal distribution.
- Rule of Thumb:
  - When $\boldsymbol{n} \geq \mathbf{30}$, the t-distribution and z-distribution are almost identical.
  - For large samples, using the z-distribution instead of t makes little practical difference.



How does using z-distribution vs. t-distribution affect the interval estimation?

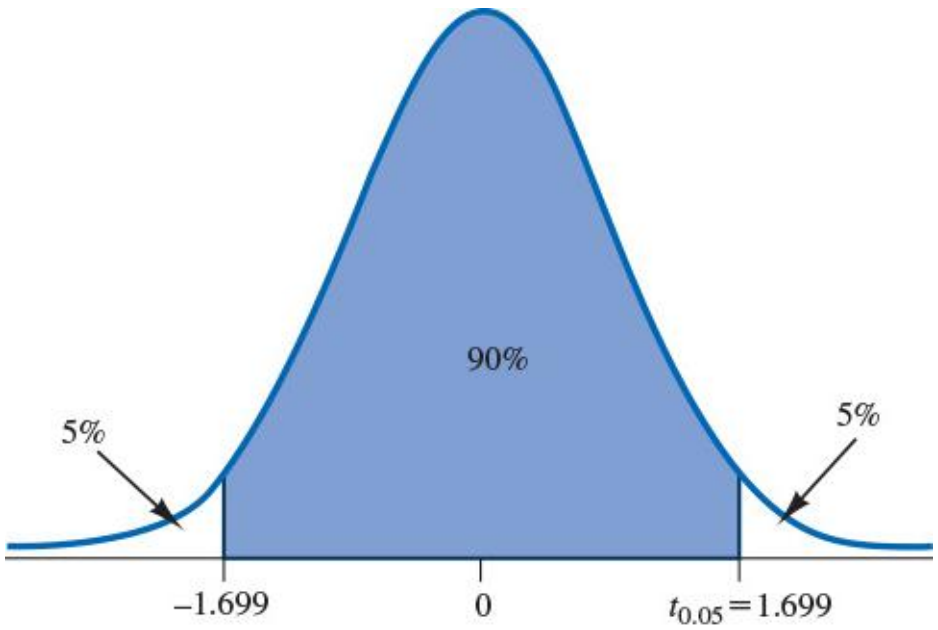
- Both use the same confidence (significance) level, but they rely on different critical values (z or t).
- The t-value is **larger than** the corresponding z-value when the sample size is small, leading to a **wider confidence interval** to **reflect greater uncertainty**.
- As the sample size increases, $t \approx z$, and the intervals become almost identical.

Interval Estimate of the Population Mean $\boldsymbol{\mu}$

- The interval estimate of the population mean μ, using the sample standard deviation s as an estimate of the population standard deviation σ, is written as:

$$\bar{x} \pm \boldsymbol{t_{\alpha/2}} \frac{\boldsymbol{s}}{\sqrt{n}}$$

- For 29 degree of freedom, we have $t_{0.05} = 1.699$, what is $z_{0.05}$? $t_{0.05/2} = 2.045$, what is $z_{0.05/2}$?
- A 95% confidence interval for the population mean (sample size $n = 30$, $\bar{x}$=71,814, $s/\sqrt{n} = 611.3$) is:

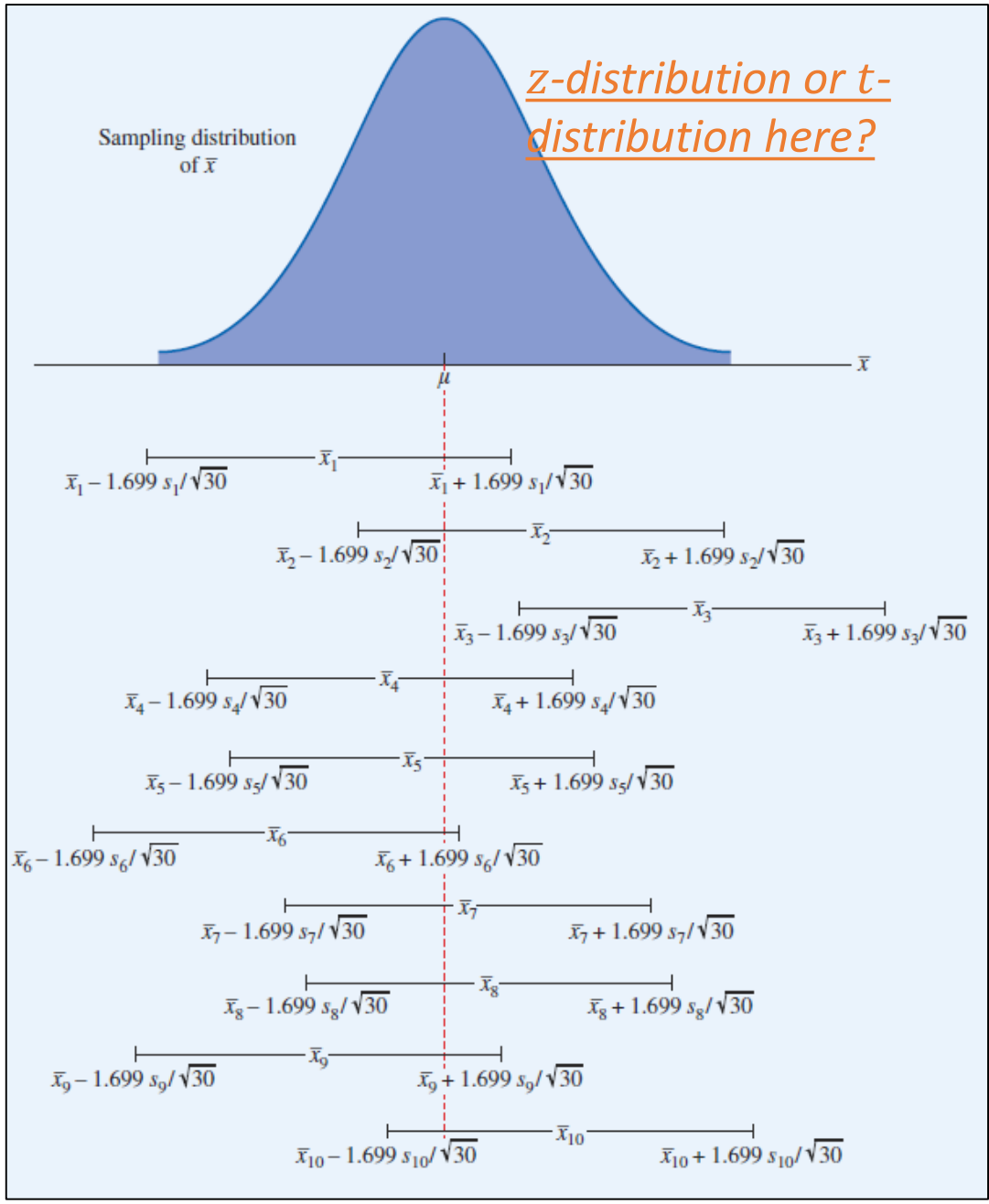


$$\bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}} = 71{,}814 \pm 2.045(611.3) = 71{,}814 \pm 1{,}250$$

<u>Question:</u> *How to explain this confidence interval?*

Interpretation of Confident Interval

- Approximately 90% of all the intervals constructed using $\bar{x} \pm 1.699(s/\sqrt{30})$ will contain the population mean.
- We are approximately 90% confident that the interval $[\$70{,}778, \$72{,}850]$ includes the population mean μ.
- This interval has been established at the 90% confidence level.
- 95% confidence interval does not have a 95% probability of containing the population mean μ.
- Once constructed, a confidence interval will either contain the population parameter or not contain the population parameter.
- If we *take several independent samples* of the same size from our population and construct a 95% confidence interval for each of these samples, we would expect *95% of these confidence intervals to contain the mean μ*.



Camm Ch7, p342, Figure 7.11

Confident Interval for a Proportion

- An unbiased estimator of a population proportion p is the statistic:

$$\hat{p} = \frac{x}{n} \text{ (the sample proportion)},$$

where

  - x is the number in the sample having the desired characteristic
  - n is the sample size

- A 100(1 – a)% confidence interval for the proportion is:

$$\hat{p} \pm z_{\alpha/2}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Sample standard deviation for $\hat{p}$ is: $\boldsymbol{\hat{p}(1-\hat{p})}$

Example: Confident Interval for a Population Proportion

- **Scenario:** A survey of 200 WKU students finds that 60 prefer mobile ordering at the campus café.
  1) Estimate the population proportion (p) of all students who prefer mobile ordering, with **95%** confidence.

- Step 1. Compute the sample proportion $\hat{p} = \frac{60}{200} = 0.30$
- Step 2. Find the standard error (SE) $SE = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \sqrt{\frac{0.3(1-0.3)}{200}} = 0.0324$
- Step 3. Identify the z-value for 95% confidence $z_{0.025} = 1.96$
- Step 4. Compute the confidence interval $\hat{p} \pm z \times SE = 0.30 \pm 1.96(0.0324)$
  $\Rightarrow$ (0.236, 0.364)

  2) What is the interpretation of this 95% confident interval?

We are **95% confident** that between **23.6% and 36.4%** of all WKU students prefer mobile ordering.

Confident Intervals and Sample Size

- We can determine the appropriate sample size needed to estimate the population parameter within **a specified level of precision (± E)**.
- Sample size for the mean:

$$n \geq (z_{\alpha/2})^2 \frac{\sigma^2}{E^2}$$

- Sample size for the proportion:

$$n \geq (z_{\alpha/2})^2 \frac{p(1-p)}{E^2}$$

[Use the sample proportion from a preliminary sample as an estimate of p or set p = 0.5 for a conservative estimate to guarantee the required precision (maximizes qty of $p(1-p)$).]

Summary: Interval Estimation

- Today's Key Points
  - A point estimate provides a single best guess for a population parameter.
  - An interval estimate gives a range of likely values:
  - Point estimate ± Margin of error
  - Confidence level reflects *how sure we are that the interval captures the true parameter*.
  - Use the z-distribution when population std σ is known (or $\boldsymbol{n \geq 30}$).
  - Use the t-distribution when population std σ is unknown and n is small.
  - As sample size increases → $t \approx z$ → **narrower** confidence intervals.
- Next Class: Hypothesis Testing
  - Logic of statistical testing: claim → evidence → decision
  - Null vs. alternative hypotheses (H_0 vs. H_1)
  - Type I and Type II errors
  - Relationship between confidence intervals and hypothesis tests